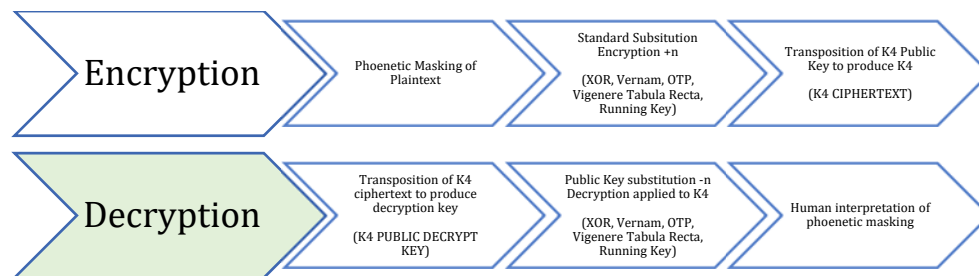


Supporting a Two-Step Decryption

October 20, 2019 – Finding Supporting Facts for the Single (Public) Key-Transposition-Substitution-Concealment Mask Decryption (identical to C2 K4 Double-Mask-OTP hypothesis 2019-07-15)

Decryption Approach from Moscow Billboard Problem (2019-10-18)



Establish the K4 Public Key with a matrix of 14 column x 7 rows (98 = 7 x 2 x 7)

?	O	B	K	R	U	O	X	O	G	H	U	L	B
S	O	L	I	F	B	B	W	F	L	R	V	Q	Q
P	R	N	G	K	S	S	O	T	W	T	Q	S	J
Q	S	S	E	K	Z	Z	W	A	T	J	K	L	U
D	I	A	W	I	N	F	B	N	Y	P	V	T	T
M	Z	F	P	K	W	G	D	K	Z	X	T	J	C
D	I	G	K	U	H	U	A	U	E	K	C	A	R

Transposition of Public Key to BERLINCLOCK crib for standard decode

								M	U	Y	K	L	G
K	O	R	N	A									

Apply XOR, Vernam, OTP, Vigenère Tabula Recta, or Running Key Decryption

								B	E	R	L	I	N
C	L	O	C	K									

1:1 Relationship is maintained as the ultimate plaintext

Proposed Transposition Method

The transposition is a stream cipher done in the analog style of a Jefferson disk. 14 disks of 7 characters are ordered, and then rotated/spun around to produce the K4 public decryption key used for the substitution decryption.



Since conventional substitution should be used, BERLINCLOCK on positions 64-73 will require K4 to be manipulated to produce MUYKLGKORNA as a partial crib. The question mark will be skipped, but was required for the original matrix.

Running the proposed Moscow Billboard through the wringer

The next goal is to see if the proposed approach can be supported through the accounts by various members in the Yahoo! Kryptos Group.

2013 Richard SantaColoma notes

In light of the fact that Sanborn had, in November of 2010, given the clue that “letters 64-69 NYPVTT in part 4 encode the text BERLIN”, several questions were asked about this. One question was, “Then you are saying that N=B, Y=E, etc.?”. Sanborn answered to the positive, while Ed seemed less ready to acknowledge this. I do not know if this is because of a misunderstanding of the question, or because of Ed’s reluctance to go that far, or because Ed himself was not entirely certain if this was the final outcome. However, as Ed professes to not know the actual answer, although he knows the cipher type used, he therefore should know whether or not there is an actual letter for letter correspondence. So I came away from the answer thinking that either Sanborn did not exactly know what was being asked, or that there is this letter for letter correspondence. I favor the latter

If the above is true, then a 1:1 relationship is possible, but Ed’s reluctance is more of a technicality whereby a transposition of K4 (Public Key) is required to decode NYPVTT to BERLIN. From Jim’s standpoint, K4 can also be treated as a single polyalphabetic substitution decryption with an encrypted transposed public key.

Ed Schiedt made the point that (paraphrasing as close as I can remember), “Panel four’s cipher was designed so that English could not be used as an aid in deciphering”. Questions were asked about this statement, among them, “Are you saying it is not in English?”, and Ed seemed reluctant to answer directly... only repeating “...English could not be used as an aid”. So my take on that is that it is either not in English, or the system employed would obviate any attempts at using character counts to solve it.

The masking system is done at the plaintext level as a phonetic mask to deter brute forcing.

2019 Richard SantaColoma notes

German Guy's Proposal: Elonka asked Jim if anyone who guessed through his proposal "system" (the \$50 a shot thing) had come close. At first, Jim said "no". But then after a pause, he said something to the effect of, "Well one guy worried me for a minute", or maybe, "One got had me scared", or like that. Elonka asked him to elaborate, and he said that a German guy had given a "solution" that seemed very close at first, but then, on (some sort of) further examination, turned out to be wrong. So this begged the question... and Elonka and I asked in different ways... "what" about the solution made him think it was wrong, then? The only thing I could think was that the result was close to correct, but that the method used told Sanborn that the result was either a lucky guess or happy error. So I asked, specifically, after a short back and forth, "If a person gave you an answer which was 100% correct, would you care how the person arrived at it?" He answered, firmly, "I don't care how they do it".

To properly decode K4 97 out of 97 characters, there is only ONE possible public key. Jim would be indifferent on the transposition process of K4 to obtain the final public key.

Transposition information

"Matrices": Jim was explaining, on at least two occasions, his passion for matrices. But he was also discussing, using hand gestures, "rotating" them, and I think (?) "flipping" or "mirroring" them (could be wrong on those last two). As a noobie to this, and not being a cipher expert, I mention it because I am not sure that "rotating" the matrices was a part of the solved K1-K3. If not, then I would say this is some sort of possible suggestion, because he was stating it in relation to these panels, perhaps a further way to glean another layer of information from them? Yeah, not strong... but again, mentioned "because".

The transposition of K4 ciphertext to the produce the key is unconventional (non-rule based).

Megellis – March 2019 Kryptos Lunch notes

Heard 2 things from Ed Scheidt. For one, he mentioned the idea of a prism as being useful. As he did not elaborate, I take it he was comparing a masking layer to a piece of glass which takes in a mixture of light wavelengths and splits them into the different components of white light. I can't say I'd ever looked at it quite that way, but it doesn't tell me anything new about what a masking layer does. It's not an all-fours metaphor, as the prism differentiates the spectrum based on the specific characteristics of the wavelength mix put into it; a masking layer, so far as I can understand it, specifically ignores the values input, and "bends the colors", so to speak, into a variegated output which has as little relation to the input as can be managed. Anyway, I did understand that statement to have been in relation to Kryptos-cracking.

Ed Scheidt offers another cryptic clue on a potential stream cipher for the transposition component of K4. This is not the first time he has mentioned the prism metaphor – which may imply a stream of SEVEN rows (Red-Orange-Yellow-Green-Blue-Indigo-Violet) instead of SEVEN colours from white light.

2019 Elonka Dunin – Langley Visit Day 3 – March 2019

When asked whether it was necessary to know the solutions to 1-3 in order to solve 4, Jim said no.

The ciphertext & plaintext of K1, K2, and K3 are not required to decrypt K4. K4 can be used to decrypt itself – literally.

It's not necessary to have a particular book to solve K4, just "to be able to read".

Referring to the plaintext being phonetically masked – likely there's a reflective or circular process of transforming K4, applying it to K4 to be decrypted and then repeating the process until words or phrases appear.

We spent quite a bit of time asking Jim about the correlation between the plaintext BERLIN and the ciphertext such as NYPVTT. Specifically, we were trying to find out if there was a 1:1 relationship from NYPVTT to BERLIN, or there was some other step, the masking technique. Jim was confused when we mentioned masking technique, evidently it's something that Ed said that Jim didn't understand. Jim said that yes the ciphertext and plaintext were connected, but when I tried to explain what exactly we were asking, like that in K1 EMUFPH is exactly BETWEE, but in K3 ENDYAH does not map exactly to SLOWLY, Jim backed off and said he would only commit to the fact that K4 is exactly 97 characters long, and that BERLIN is plaintext that starts at exactly the 64th character, but he wouldn't go further than that.

If K4 is decrypted by transposition (to produce public decrypt key) then substitution, Jim would prefer not to disclose the steps to decoding K4. BERLIN starting exactly at the 64th position means that there is some form of substitution occurring from the 64th to 73rd position to produce BERLINCLOCK. The transposition portion is not exactly manipulating the ciphertext, but assembling the public decryption key.

2015 Scheidt Workshop Transcript – Richard Bean

[3:55] I look at cryptography as building things ... there's other aspects, and maybe from your perspective ... in the sense of ... if you had a cryptographic paradigm is there a way to break it or understand it or what's the schema that's involved in that.

Classical ciphers are fundamentally substitutions and transpositions.

Q: I've got K1, 2 and 3 are one-stage encryption - is that also true for K4 [47:06]
aha - now yeah you're closer to that answer than [saying??] is at one stage [47:16]
no, I would consider it more than one stage [47:19]

The proposed decryption is a two-stage process.

I've mentioned earlier and I have mentioned in public there's a masking technique that applied so that the English language wouldn't be available [47:35]

Masking to be done at the plain text level in the event the approach is discovered.

[what is clue in plain sight / duress code] [53:45] [Jim] has said that something that was key is no longer there. Jim questions ... [no clear answers]
[masking first then encryption? or the other way around] [no clear answer]

Masking is a destructive/irreversible step on the plaintext – it can ONLY be done on the plaintext.

Bill Briere Notes – Nov 3, 2013

8. Someone asked about the previously stated one-to-one correspondence of the plaintext to the ciphertext, at least for positions 64-69. Many accept this as settled; but I don't think it is. It's not that I'm clinging to the need for it to NOT be that, but I think that some of the likely candidates would require it. Jim again stated, pretty convincingly, that it was 1:1. I asked Ed -- some might say untactfully -- if he would answer the same question. He hesitated and looked to Jim for guidance on what to say, then didn't say anything. I believe that Ed's hesitation indicates that either it was NOT a 1:1 correlation, or that the answer was not something that was black-and-white, or that he could not answer without maybe taking a look at the piece. Again, Ed may not know exactly what was done, but he might be skeptical of what Jim thinks he knows about 1:1. My take-away is this: If Ed's answer were going to be the same as Jim's, there wouldn't have been any hesitation in answering. Also noted, a bit later (I think),

that Jim gave Ed (and maybe me, too) a dirty look, possibly for talking too much, and indicated that we had gotten quite enough information already and that he (Jim) would be the only one to give out clues. (Sorry, Jim. Love you, man.)

Ultimately K4 is decrypted from a standard polyalphabetic substitution, BUT the key must be constructed by changing the character positioning of K4 to obtain the K4 public key. This technicality is what makes Ed hesitate because it could severely mislead Kryptos codebreakers.

Thomas Fulmer – 2007 DC Visit

3. Question was asked if he considered substitution followed by transposition (or the reverse) qualified as "masking" --Ed Laughed and gave no reply

Masking is done at the plaintext level, and if Ed answered, he would be giving away the decryption steps.

4. In a conversation with Elonka, Ed repeatedly spoke about the code being "Reflective" When asked if he meant symmetric, he replied, "Reflective"

Reflective describes the circular process where K4 is transformed and applied to K4 (decryption) to produce a plaintext result. Symmetric would imply that the K4 public key is the same for encryption and decryption. In terms of standard XOR, Vernam, OTP, Vigenere, or running key, the keys are asymmetric in nature. Encoding is generally a +n step, and decoding is a -n step.

Elonka Dunin – 2007 DC Visit (Dinner)

* K4 uses a system "unknown to anyone on the planet"

The unknown to anyone on the planet refers to an unorthodox transposition to produce the K4 public key.

* His job wasn't breaking codes, it was designing them

Code design deals with creating a cipher and addressing the key in terms of distribution and security.

* When I asked if the method for K4, once understood, could be re-used for other messages, he said, "The concept could."

* He says all crypto boils down to substitution and transposition (I pointed out concealment)

Conceptually a self-decoding cipher can be reused as long as the basic substitution/transposition ciphers are understood.

* He said the first parts of Kryptos were "simple" substitution, to show that it was in English. He said he could have double-transposed part 1 and it "never would have been solved"

A double transposition will equate to a single transposition. The difficulty lies with the distribution of the key and/or deciphering tableau for substitution.

* He was describing my analysis as "trying to find the method." With the implication that that's the wrong way to go about it.

K4 does not use a specific known cipher (i.e. playfair, railfence, etc.) but has its own unique analog encryption. It is more important to understand the approach first then go for the problem. In this case, it's about dealing with the transposition of K4 to produce the public key. The substitution component is largely dependent on the transposition anyways.

* Referred to "bias" as helping out the frequency counters. The masking technique is designed to remove the bias

Frequency counting would give clues on the nature of the substitution/transposition – K4 is to be decoded by analog means and not computer brute forcing.

2005 DC Trip - D1stancerunner20

We talked quite a bit about problems in high performance computing, to communications methods, and short comings in current cryptographic systems. In this discussion a couple of interesting things came up, they were not related to talking directly about Kryptos but they may have some meaning. One thing that was mentioned several times was symmetry in nature and it's value in cryptography. One thing he mentioned was how the Egyptians built the pyramids and they have standed the test of time, because of their symmetry. The concept that nature tends to be symmetrical was mentioned a couple of times.

K4 is an asymmetrical cipher in the sense that the K4 public decryption key is different from the encryption key. In this sense, this is Ed Scheidt cryptically referring to public key cryptographic methods.

Another thing that I cued in on was when he mentioned light and it's use in cryptography. This was not when talking about Kryptos either. However, he mentioned about the use of light in cryptography. He mentioned a prism and how it breaks white light into it's individual components. Then came the part that fascinating; when talking about electromagnetic signals (this part is theoretical), he mentioned modifying either the H vector or the E vector, and being able to transfer light information onto twisted-pair (copper wire). He said he presented this idea about 5 years ago so it may not have any relation to Kryptos (although he may have been thinking about it for some time). Interesting, Kryptos is copper and conveys information through light by us being able to see it in the visible spectrum :)

Prism referring to a potential version of a stream cipher for the transposition component. A prism may refer to 7 streams from the 7 colours.

2005 Elonka Dunin – 2005 DC Notes

Scheidt said that every letter in K4 will be used, "like the other sections"

Refers to the usage of all 97 characters. The question mark remains in limbo, but has been used for the hypothesized decryption.

Scheidt said he has not done a reverse check of K4 himself, but he's confident that it was done correctly.

Since it is a two step process, Ed can confirm at the K4 public key level (only one possible key) and not apply the decryption steps to K4.

2013 Kryptos Dinner - Doug Price

3. Ed implied (in fact, I'd go as far as saying he "stated") that there is a masking step prior to the encryption. K1/2/3 have English as the plaintext, where you can use language statistics to help solve the cipher. "In K4, I took that advantage away from you". He implied/stated that he did something with the plaintext prior to encrypting to change the language statistics; this is different than just using a long key (running key, composite key, etc.) to encrypt the message -- if he did that he would not have described his process the way he did.

This is the only time that Ed has stated the masking step is done PRIOR to encryption. That is, the masking step is done on the PLAINTEXT.

2006 New York Times Interview – Ed Scheidt

He (Ed Scheidt) said a good code, even if it's complex, should be usable easily.

The proposed decryption is not complex and easily usable (conventional substitution using a K4 public key)

2015 Ed Scheidt Workshop – Elonka Dunin Notes

When asked about whether the masking technique was applied to the plaintext before encryption, or after decryption, he "deferred".

Refer to 2013 Kryptos Dinner – Ed says masking step prior to encryption.

2013-05-30 <https://kryptosfan.wordpress.com/2013/05/30/jim-sanborn-answers-kryptos-questions/>

Is the 4th part 97 or 98 letters? try both

98 characters are required for the transposition of K4 (step one to produce the public decryption key)

Who do you think is most likely to solve K4? a Vulcan

KRYPTOS must be deciphered in a logical manner using factual information.

Is the actual code/cipher breaking secondary to the actual process of understanding what Kryptos really means? both are important

Without understanding of the cipher, the plaintext does not have as much meaning.

Any advice for people who “get so close” after a complicated multi-step decryption effort who are convinced that they just need more steps or help to solve it? you get it all or nothing at all

The K4 Public Key must be transformed perfectly.

2013 ACA Convention: - Jim M's notes

It was asked if N decodes to B, Y decodes to E, etc, etc. and Jim confirmed it does. Emphatically.

Jim Sanborn rattled through the entire crib:

N = B,

Y = E,

P = R,

V = L,

T = I,

T = N

When the same question was asked to Ed, Jim cut in and said “I’m the only one who knows.”

It was asked if N decodes to B or encodes to B, and Jim stated “decodes”.

It was asked if K4 was a “one-off”. Some discussion ensued about the definition of “one-off”, but Ed looked to Jim and said “well Jim, how are we going to answer that one?”. To which Jim simply replied “we don’t answer it”.

The most important information is that K4 is done by decryption. This is very important for standard substitution ciphers (XOR, Vernam, OTP, Vigenère Tabula Recta, etc.)

Elonka jumped in to remind Ed that he once told her that Kryptos uses a system unknown to anyone on the planet. Ed stated he didn’t recall that conversation, and back-pedaled out of saying anything more.

The unknown part refers to the unique analog process for transposition.

1991 ABC Interview

How did you decide what message to encode?

Well that changed during the course of the project as well. And that developed largely from the reading that I did about the Agency. And also while I was cutting out, physically cutting out the copperplates, things in my own mind changed. I wanted to put something on there which would -- Let’s just say, that once the plate is deciphered, I’m not convinced the true meaning will be clear even then. I’ve made a statement which is straightforward, but that leads to something else. There’s another deeper mystery. As you peel off a layer of an onion, the myst-- you get closer to the heart of what it is. And so y’know I just wanted to make it even after it was deciphered, you had to go deeper and decipher something else. It’s in English, it’s in plain English, it’s in text, and you can read it, but that isn’t necessarily the whole story.

Implications of a potential K5.

2011 CBC Interview w/ Q Podcast: Jian Gimeshi for BERLIN clue – Jim Sanborn

54:31 **Interviewer:** Which I certainly don’t know wh-what that means, other than one of my favorite cities in the world, but...ah...Do you expect that we’ll see the code broken anytime soon?

54:39 **JS:** It’s wha- It 6 of 97 characters. The odds...the odds of getting the rest--I mean if you’re a very good cryptographer, your odds are certainly improved. But the odds of-of-of filling in the lines in text around it, you know, are-it’s still difficult and very challenging code, I must say.

And...um...That’s not to say won’t give another-, in another decade, when I am 75 [Interviewer laughs] I might give another clue. And maybe when I am 85, I will give another clue. And all these clues, perhaps, will add up to something that, you know, is-is infinitely crack-able. Um...I started this piece the in 1988, and installed it in like 1990. And...um...During that period of time, the Berlin Wall fell.

The infinite portion refers to the trial and error of the K4 public key to produce a plaintext. Since it appears the K4 public key transformation algorithm is not disclosed, this may suggest the infinite possibilities.

55:29 **Interviewer:** Well, when this [Break] Well, the...Maybe you just gave us another clue!? I’m..people [SB Laughs] ju..I hope that people are listening to this. That...Ah...um...That, in fact, everything that you say is a clue, I-I am even convinced.

55:39 **JS:** Well, yes, there is--there is a serious, [Interviewer laughs] sort of, forensic..um..[SB laughs] [Interviewer laughs] dissection ...

Berlin itself is extremely important as the K4 decryption would require MUYKLG decryption of NYPVTT to produce BERLIN. There is only one M and one Y in K4.

2003 Sanborn Trip – Elonka Dunin

- He commented how it was odd that "no one has recovered the original matrix". He kept using that word "matrix" quite a bit, such as to say "matrix system". Evidently there's something important about re-creating the exact system that he used for encrypting the messages, and he has never seen anyone do that yet (and didn't see it anywhere on my slides).

Matrix may refer to the 14x7 K4 matrix that will be transposed to yield the K4 public decryption key.

- Similar to how Scheidt reacted, I saw Sanborn light up when I talked about how the extra "L" on the tableau side now means that we have the same number of characters on both the cipher alphabet and tableau sides. Neither Scheidt nor Sanborn *said* anything when I talked about that, but I definitely noticed both of them reacting, like sitting up or forward a little more, nodding like this was something that they knew and were glad that someone had noticed.

The extra L refers to the stenographic clue for K2 deciphering tableau. This outlines the tableau (top right portion) and the bottom left is found by looking at the W in the tableau left side.

- When I asked about the misspellings and asked if they were accidental or deliberate, Sanborn said that they were deliberate, but it was less important *what* they were. He said, and I quote: "it's more the orientation of those letters that's useful there." Later on in the evening he repeated that point, saying it was the "positioning" that was important.

Perhaps the errors may suggest a hidden algorithm such as PALIMP(C)EST being the 7th character on the key.

- Sanborn was adamant that Scheidt did *not* do the encryption. He said that Scheidt did teach him various systems "going back to Caesar", and that Sanborn used "several" of them, but that the actual encryption was done by Sanborn, by himself, during the time that he was driving the petrified tree from the southwestern U.S. back to the DC area. Sanborn said that he and Scheidt had agreed to keep it "compartmentalized" like that. Sanborn insisted that when it came to the encryption, he said (referring to himself) that it was "entirely me."

The encryption steps are fairly straightforward: write out plaintext, phonetically mask the plaintext, encrypt +n and then scatter (transform) the result.

- Sanborn said that no one else beta-tested the code -- not even Scheidt. When I commented that first-time code creators often make errors that leave a code unsolvable, he said that he was certain that he did it right. He said though he has never gone back to decrypt it, because that keeps it more secure.

There is ultimately one combination on K4 public key that will yield 97 of 97 characters for the decrypt.

- Getting back to Kryptos, Sanborn commented that he was surprised that no one had tried recovering the original matrix and running it through all possible "shifts".

- He said that "Kryptos wasn't cracked the reverse of the way that I did it."

Refers to a potential asymmetrical public key problem where the encryption steps will differ from the decryption steps.

2010 – NPR All Things Considered – Radio Interview w/ Jim Sanborn

JS: OK, well I mean really, it's 6 characters out of 97. And I dangled the clue BERLIN, but I also divulged or gave images of my original decoding charts, the ones that I... well actually for me they were encoding charts. And I think once the Kryptophiles study it in a forensic manner there might be revelations in there. So in a way I gave more than just BERLIN, I think I gave other information as well.

BERLIN is a vital clue in the process due to the required conventional substitution of NYPVTT being decoded by MUYKLG. There is only 1 M and 1 Y present on K4 and they are not on the 64th and 66th positions respectively. Some sort of transposition IS required to construct the K4 public key.

2014 – New York Times Clue (CLOCK)

When asked whether his new clue was a reference to this Berlin clock, Mr. Sanborn, sounding pleased, said, "There are several really interesting clocks in Berlin."

He added, "You'd better delve into that particular clock," a favorite of conspiracy theorists because of the mysterious death in 1991 of its designer, Dieter Binninger. With all the intriguing timekeepers in the city, including the "Clock of Flowing Time," Mr. Sanborn said, "There's a lot of fodder there."

The particular clock may refer to the BERLINCLOCK crib literally. To decrypt NYPVTMZFPK, the characters MUYKLGKORNA are required and can be constructed from the transformation of K4 into a public key.

Conclusion

The Moscow Billboard approach is identical to the C2 K4 Double Mask-OTP Hypothesis proposed on July 15, 2019. Through review of information gathered by the Kryptos community over the years, the approach can be substantiated.